Project 3 - Unsupervised learning:

Employee Performance Analysis : Employee Performance Analysis:

```
    Analyze employee performance data to identify performance patterns or cluste
rs. Procedure: Apply K-means on employee performance metrics, such as productivi
ty, attendance, feedback scores, or goal achievements. Domain: Human Resources
```

In [20]:

```
import numpy as np
import pandas as pd
data2 = pd.read_csv('Train.csv')
data2
```

Out[20]:

ss	Time_of_service	Time_since_promotion	...	Compensation_and_Benefits	Work_Life_balance
ıal	4.0	4	...	type2	3.0
:al	5.0	4	...	type2	4.0
ıal	27.0	3	...	type2	1.0
ral	4.0	3	...	type2	1.0
ıal	5.0	4	...	type3	3.0
...	...	...	...	...	...
ral	4.0	1	...	type3	3.0
:al	16.0	2	...	type3	3.0
ve	10.0	1	...	type2	4.0
ral	24.0	1	...	type0	2.0
ve	22.0	2	...	type2	3.0

In [21]:

```
data2.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7000 entries, 0 to 6999
Data columns (total 24 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   Employee_ID                7000 non-null   object
 1   Gender                     7000 non-null   object
 2   Age                        6588 non-null   float64
 3   Education_Level            7000 non-null   int64
 4   Relationship_Status        7000 non-null   object
 5   Hometown                   7000 non-null   object
 6   Unit                       7000 non-null   object
 7   Decision_skill_possess     7000 non-null   object
 8   Time_of_service            6856 non-null   float64
 9   Time_since_promotion       7000 non-null   int64
 10  growth_rate                7000 non-null   int64
 11  Travel_Rate                7000 non-null   int64
 12  Post_Level                 7000 non-null   int64
 13  Pay_Scale                  6991 non-null   float64
 14  Compensation_and_Benefits  7000 non-null   object
 15  Work_Life_balance          6989 non-null   float64
 16  VAR1                       7000 non-null   int64
 17  VAR2                       6423 non-null   float64
 18  VAR3                       7000 non-null   float64
 19  VAR4                       6344 non-null   float64
 20  VAR5                       7000 non-null   int64
 21  VAR6                       7000 non-null   int64
 22  VAR7                       7000 non-null   int64
 23  Attrition_rate             7000 non-null   float64
dtypes: float64(8), int64(9), object(7)
memory usage: 1.3+ MB
```

In [22]:

```
data2.head()
```

Out[22]:

ss	Time_of_service	Time_since_promotion	...	Compensation_and_Benefits	Work_Life_balance
ıal	4.0	4	...	type2	3.0
:al	5.0	4	...	type2	4.0
ıal	27.0	3	...	type2	1.0
'al	4.0	3	...	type2	1.0
ıal	5.0	4	...	type3	3.0

In [23]:

```
data2.tail()
```

Out[23]:

	Employee_ID	Gender	Age	Education_Level	Relationship_Status	Hometown	Un
6995	EID_16328	F	23.0	5	Married	Franklin	Operarion
6996	EID_8387	F	44.0	1	Married	Lebanon	R&
6997	EID_8077	F	49.0	3	Single	Springfield	I
6998	EID_19597	F	47.0	3	Married	Washington	Sale
6999	EID_1640	F	58.0	3	Married	Franklin	I

5 rows × 24 columns

In [24]:

```
data2.describe()
```

Out[24]:

	Age	Education_Level	Time_of_service	Time_since_promotion	growth_rate	Ti
count	6588.000000	7000.000000	6856.000000	7000.000000	7000.000000	70
mean	39.622799	3.187857	13.385064	2.367143	47.064286	
std	13.606920	1.065102	10.364188	1.149395	15.761406	
min	19.000000	1.000000	0.000000	0.000000	20.000000	
25%	27.000000	3.000000	5.000000	1.000000	33.000000	
50%	37.000000	3.000000	10.000000	2.000000	47.000000	
75%	52.000000	4.000000	21.000000	3.000000	61.000000	
max	65.000000	5.000000	43.000000	4.000000	74.000000	

In [25]:

```
data2.dtypes
```

Out[25]:

```
Employee_ID                   object
Gender                        object
Age                          float64
Education_Level                int64
Relationship_Status           object
Hometown                      object
Unit                          object
Decision_skill_possess        object
Time_of_service              float64
Time_since_promotion           int64
growth_rate                    int64
Travel_Rate                    int64
Post_Level                     int64
Pay_Scale                    float64
Compensation_and_Benefits     object
Work_Life_balance            float64
VAR1                           int64
VAR2                         float64
VAR3                         float64
VAR4                         float64
VAR5                           int64
VAR6                           int64
VAR7                           int64
Attrition_rate               float64
dtype: object
```

In [26]:

```
data2.isnull().sum()
```

Out[26]:

```
Employee_ID                   0
Gender                        0
Age                         412
Education_Level               0
Relationship_Status           0
Hometown                      0
Unit                          0
Decision_skill_possess        0
Time_of_service             144
Time_since_promotion          0
growth_rate                   0
Travel_Rate                   0
Post_Level                    0
Pay_Scale                     9
Compensation_and_Benefits     0
Work_Life_balance            11
VAR1                          0
VAR2                        577
VAR3                          0
VAR4                        656
VAR5                          0
VAR6                          0
VAR7                          0
Attrition_rate                0
dtype: int64
```

In [27]:

```
data2.dropna(inplace = True)
```

In [28]:

```
data2
```

Out[28]:

	Employee_ID	Gender	Age	Education_Level	Relationship_Status	Hometown	Un
0	EID_23371	F	42.0	4	Married	Franklin	I
1	EID_18000	M	24.0	3	Single	Springfield	Logistic
2	EID_3891	F	58.0	3	Married	Clinton	Quali
4	EID_22534	F	31.0	1	Married	Springfield	Logistic
5	EID_2278	M	54.0	3	Married	Lebanon	Purchasin
...	...	...	...	...	...	...	.
6995	EID_16328	F	23.0	5	Married	Franklin	Operarior
6996	EID_8387	F	44.0	1	Married	Lebanon	R&
6997	EID_8077	F	49.0	3	Single	Springfield	I
6998	EID_19597	F	47.0	3	Married	Washington	Sale
6999	EID_1640	F	58.0	3	Married	Franklin	I

5353 rows × 24 columns

In [29]:

```
data2.isna().sum()
```

Out[29]:

```
Employee_ID                  0
Gender                       0
Age                          0
Education_Level              0
Relationship_Status          0
Hometown                     0
Unit                         0
Decision_skill_possess       0
Time_of_service              0
Time_since_promotion         0
growth_rate                  0
Travel_Rate                  0
Post_Level                   0
Pay_Scale                    0
Compensation_and_Benefits    0
Work_Life_balance            0
VAR1                         0
VAR2                         0
VAR3                         0
VAR4                         0
VAR5                         0
VAR6                         0
VAR7                         0
Attrition_rate               0
dtype: int64
```

In [36]:

```
x = data2.drop("Employee_ID",axis='columns')
x
```

Out[36]:

ervice	Time_since_promotion	growth_rate	...	Compensation_and_Benefits	Work_Life_balance
4.0	4	33	...	type2	3.0
5.0	4	36	...	type2	4.0
27.0	3	51	...	type2	1.0
5.0	4	62	...	type3	3.0
19.0	1	21	...	type2	1.0
...	...	...	...	...	...
4.0	1	20	...	type3	3.0
16.0	2	59	...	type3	3.0
10.0	1	59	...	type2	4.0
24.0	1	21	...	type0	2.0
22.0	2	74	...	type2	3.0

In [37]:

```
data2
```

Out[37]:

	Employee_ID	Gender	Age	Education_Level	Relationship_Status	Hometown	Un
0	EID_23371	F	42.0	4	Married	Franklin	I
1	EID_18000	M	24.0	3	Single	Springfield	Logistic
2	EID_3891	F	58.0	3	Married	Clinton	Quali
4	EID_22534	F	31.0	1	Married	Springfield	Logistic
5	EID_2278	M	54.0	3	Married	Lebanon	Purchasin
...	...	...	...	...	...	...	.
6995	EID_16328	F	23.0	5	Married	Franklin	Operarior
6996	EID_8387	F	44.0	1	Married	Lebanon	R&
6997	EID_8077	F	49.0	3	Single	Springfield	I
6998	EID_19597	F	47.0	3	Married	Washington	Sale
6999	EID_1640	F	58.0	3	Married	Franklin	I

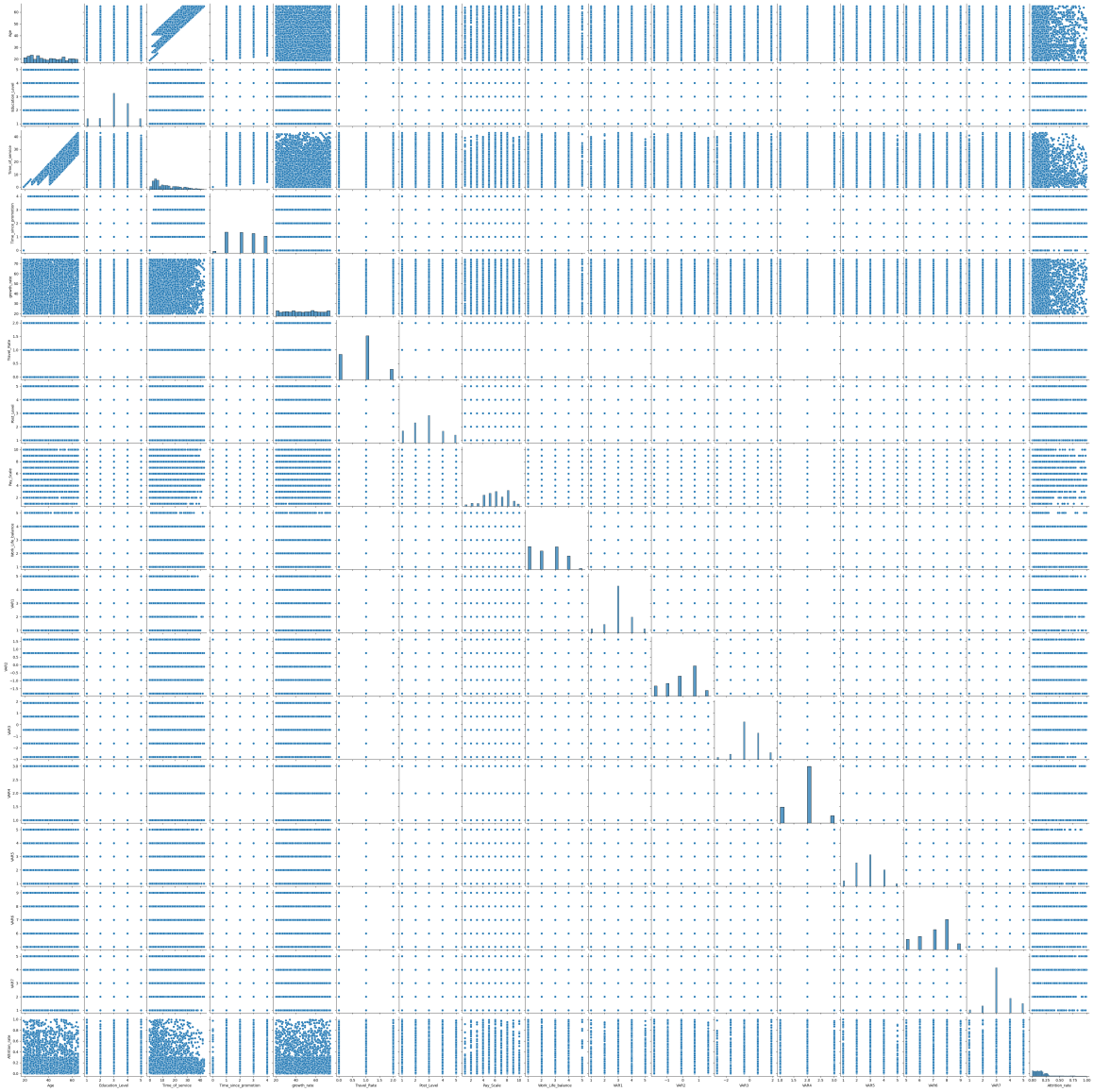
5353 rows × 24 columns

In [31]:

```
import seaborn as sns
sns.pairplot(data2)
```

Out[31]:

```
<seaborn.axisgrid.PairGrid at 0x2022d485b80>
```



In [39]:

```
#### Import label encoder
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
data2.columns
```

Out[39]:

```
Index(['Employee_ID', 'Gender', 'Age', 'Education_Level',
       'Relationship_Status', 'Hometown', 'Unit', 'Decision_skill_posses
s',
       'Time_of_service', 'Time_since_promotion', 'growth_rate', 'Travel_R
ate',
       'Post_Level', 'Pay_Scale', 'Compensation_and_Benefits',
       'Work_Life_balance', 'VAR1', 'VAR2', 'VAR3', 'VAR4', 'VAR5', 'VAR
6',
       'VAR7', 'Attrition_rate'],
      dtype='object')
```

In [42]:

```
data2['Gender'] = le.fit_transform(data2['Gender'])
data2['Relationship_Status'] = le.fit_transform(data2['Relationship_Status'])
data2['Hometown'] = le.fit_transform(data2['Hometown'])
data2['Employee_ID'] = le.fit_transform(data2['Employee_ID'])
data2['Unit'] = le.fit_transform(data2['Unit'])
data2['Decision_skill_possess'] = le.fit_transform(data2['Decision_skill_possess'])
data2['Compensation_and_Benefits'] = le.fit_transform(data2['Compensation_and_Benefits'])
```

In [43]:

```
data2
```

Out[43]:

ss	Time_of_service	Time_since_promotion	...	Compensation_and_Benefits	Work_Life_balance
2	4.0	4	...	2	3.0
0	5.0	4	...	2	4.0
2	27.0	3	...	2	1.0
2	5.0	4	...	3	3.0
2	19.0	1	...	2	1.0
...	...	...	...	...	...
1	4.0	1	...	3	3.0
0	16.0	2	...	3	3.0
3	10.0	1	...	2	4.0
1	24.0	1	...	0	2.0
3	22.0	2	...	2	3.0

In [34]:

```
from sklearn.cluster import KMeans
```

In [44]:

```
cluster = []
for i in range(1,11):
  kmeans = KMeans(n_clusters=i, random_state = 678)
  kmeans.fit(data2)
  cluster.append((i, kmeans.inertia_))
```

In [45]:

```
cluster
```

Out[45]:

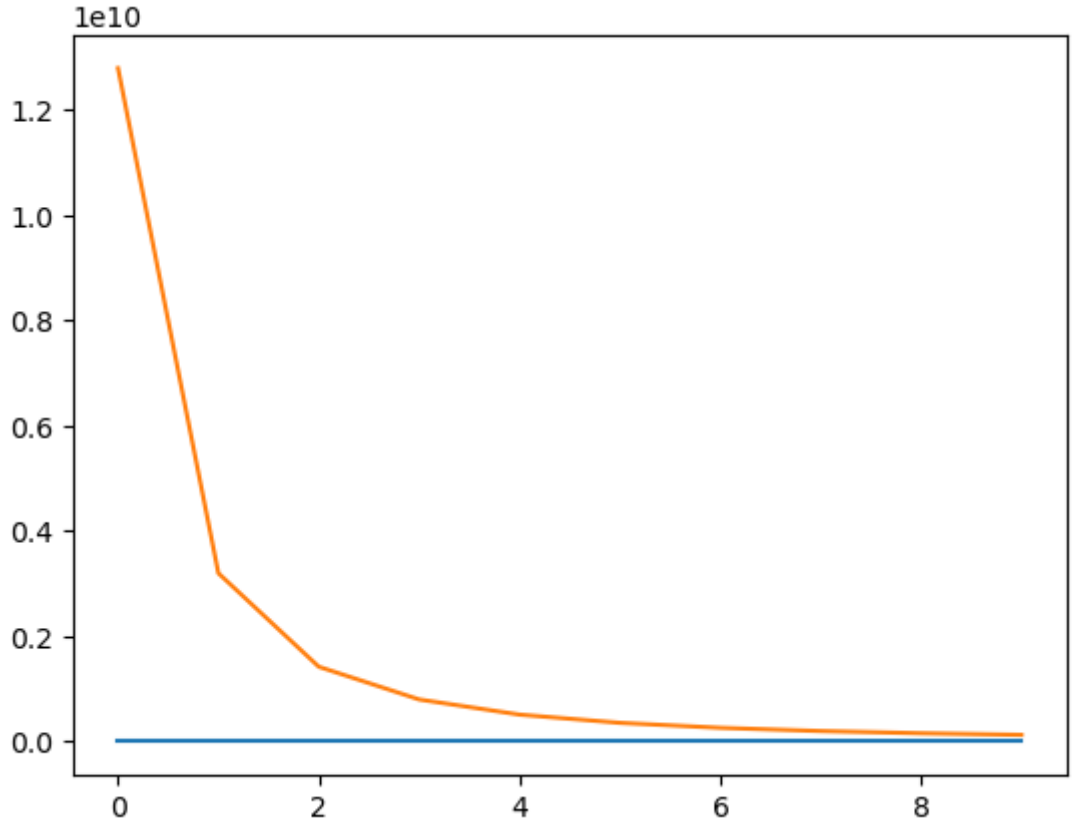
```
[(1, 12785401329.61976),
 (2, 3198301418.652826),
 (3, 1422889206.6272168),
 (4, 801507238.0148523),
 (5, 513898327.56296366),
 (6, 357678796.4127675),
 (7, 263356116.05584753),
 (8, 202276213.83294332),
 (9, 160346535.89335412),
 (10, 130307470.84125079)]
```

In [46]:

```
plt.plot(cluster)
```

Out[46]:

```
[<matplotlib.lines.Line2D at 0x2023e40cdc0>,
 <matplotlib.lines.Line2D at 0x2023e40c2e0>]
```



In [48]:

```
kmeans = KMeans(n_clusters = 5, random_state = 45)

kmeans.fit(data2)
```

Out[48]:

```
KMeans(n_clusters=5, random_state=45)
```

In [50]:

```
kmeans.predict([data2.iloc[5, :]])
```

```
C:\Users\admin\anaconda3\lib\site-packages\sklearn\base.py:450: UserWarnin
g: X does not have valid feature names, but KMeans was fitted with feature
names
  warnings.warn(
```

Out[50]:

```
array([3])
```

In [52]:

```
data2.iloc[5, :]
```

Out[52]:

```
Employee_ID                   2026.0000
Gender                           0.0000
Age                             21.0000
Education_Level                  4.0000
Relationship_Status              0.0000
Hometown                         3.0000
Unit                             7.0000
Decision_skill_possess           3.0000
Time_of_service                  2.0000
Time_since_promotion             1.0000
growth_rate                     50.0000
Travel_Rate                      1.0000
Post_Level                       1.0000
Pay_Scale                        6.0000
Compensation_and_Benefits        2.0000
Work_Life_balance                2.0000
VAR1                             3.0000
VAR2                            -0.9612
VAR3                             0.7075
VAR4                             2.0000
VAR5                             3.0000
VAR6                             7.0000
VAR7                             3.0000
Attrition_rate                   0.2819
Name: 6, dtype: float64
```

In [53]:

```
labels = kmeans.labels_
```

In [54]:

```
labels
```

Out[54]:

```
array([4, 3, 2, ..., 1, 4, 3])
```

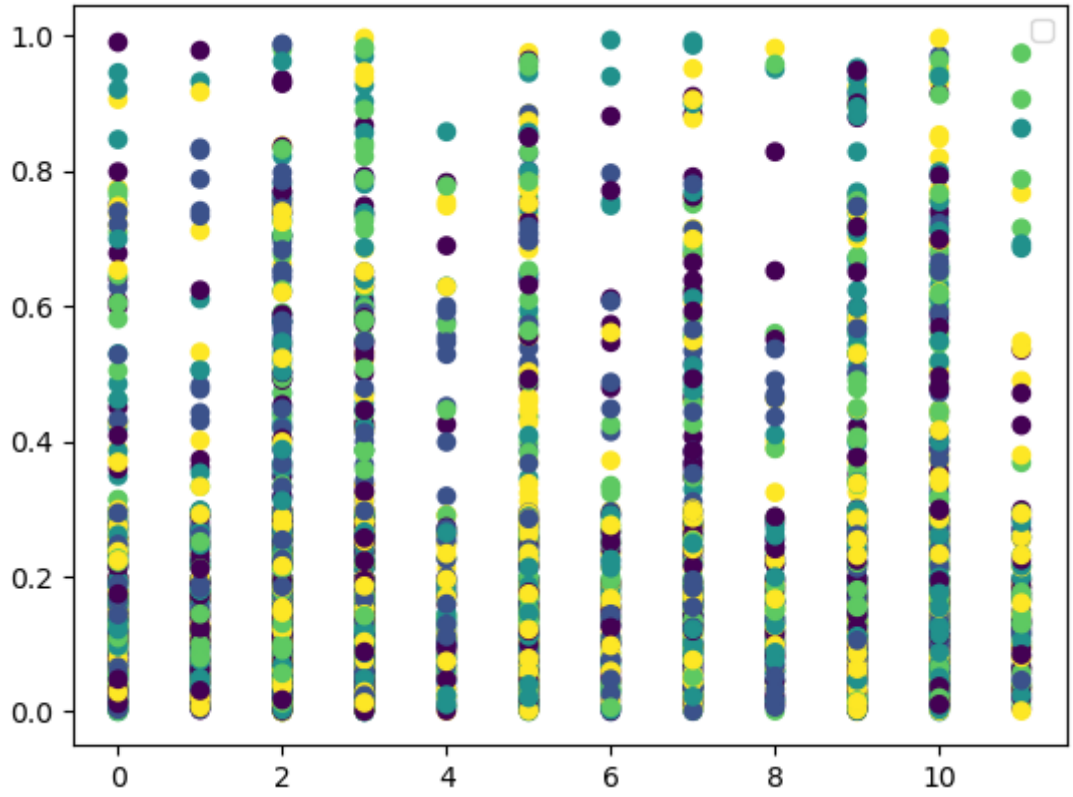
In [59]:

```
plt.scatter(data2["Unit"],data2["Attrition_rate"],c = labels)
plt.legend()
```

```
No artists with labels found to put in legend.  Note that artists whose la
bel start with an underscore are ignored when legend() is called with no a
rgument.
```

Out[59]:

```
<matplotlib.legend.Legend at 0x20242fe07c0>
```



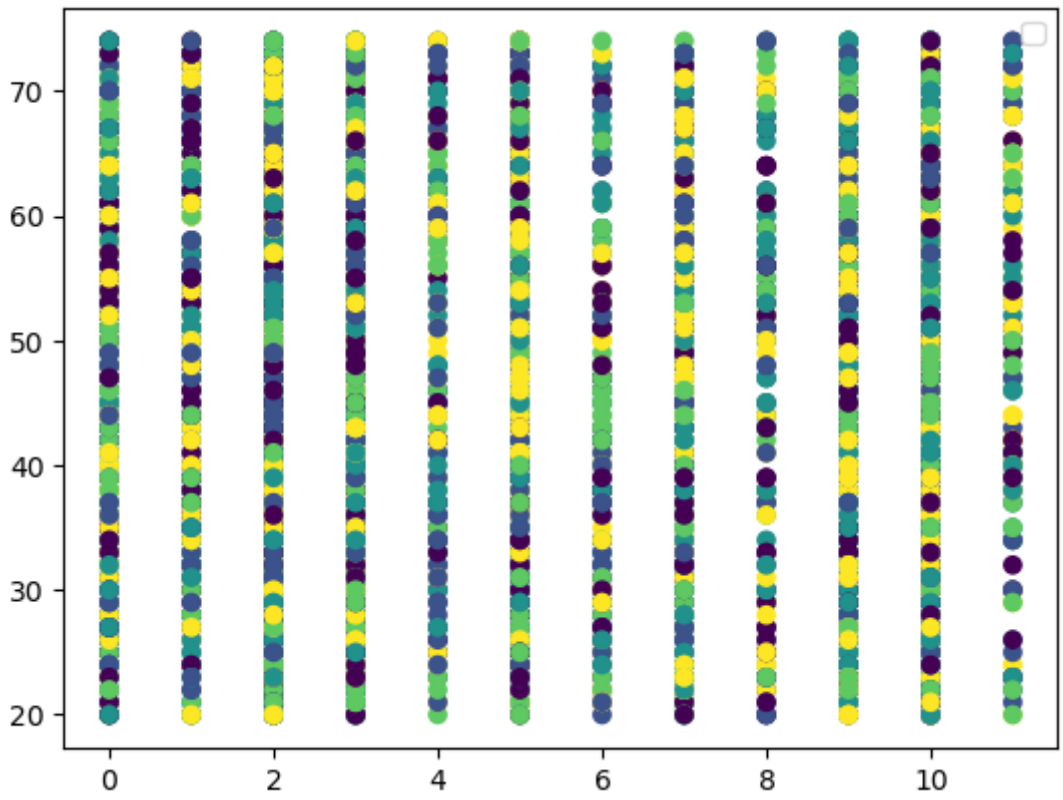
In [57]:

```
plt.scatter(data2["Unit"],data2["growth_rate"], c = labels)
plt.legend()
```

```
No artists with labels found to put in legend.  Note that artists whose la
bel start with an underscore are ignored when legend() is called with no a
rgument.
```

Out[57]:

```
<matplotlib.legend.Legend at 0x202429adbe0>
```



In [60]:

```
from scipy.cluster.hierarchy import dendrogram, linkage
```

In [62]:

```
lin = linkage(data2[["Unit","growth_rate"]][:10])
```

In [63]:

```
lin
```

Out[63]:

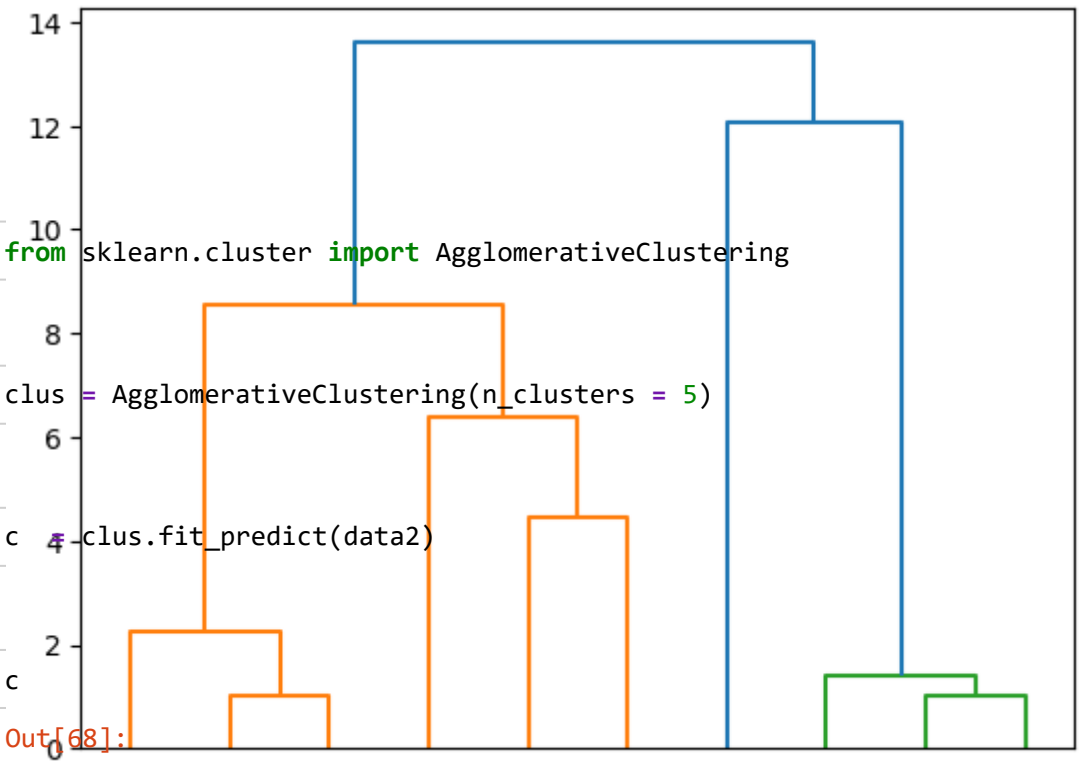
```
array([[ 0.        ,  6.        ,  1.        ,  2.        ],
       [ 5.        ,  7.        ,  1.        ,  2.        ],
       [ 2.        , 11.        ,  1.41421356,  3.        ],
       [ 1.        , 10.        ,  2.23606798,  3.        ],
       [ 4.        ,  8.        ,  4.47213595,  2.        ],
       [ 9.        , 14.        ,  6.40312424,  3.        ],
       [13.        , 15.        ,  8.54400375,  6.        ],
       [ 3.        , 12.        , 12.08304597,  4.        ],
       [16.        , 17.        , 13.60147051, 10.        ]])
```

In [64]:

```
dendrogram(lin)
```

Out[64]:

```
{'icoord': [[15.0, 15.0, 25.0, 25.0],
  [5.0, 5.0, 20.0, 20.0],
  [45.0, 45.0, 55.0, 55.0],
  [35.0, 35.0, 50.0, 50.0],
  [12.5, 12.5, 42.5, 42.5],
  [85.0, 85.0, 95.0, 95.0],
  [75.0, 75.0, 90.0, 90.0],
  [65.0, 65.0, 82.5, 82.5],
  [27.5, 27.5, 73.75, 73.75]],
 'dcoord': [[0.0, 1.0, 1.0, 0.0],
  [0.0, 2.23606797749979, 2.23606797749979, 1.0],
  [0.0, 4.47213595499958, 4.47213595499958, 0.0],
  [0.0, 6.4031242374328485, 6.4031242374328485, 4.47213595499958],
  [2.23606797749979, 8.54400374531753, 8.54400374531753, 6.403124237432848
5],
  [0.0, 1.0, 1.0, 0.0],
  [0.0, 1.4142135623730951, 1.4142135623730951, 1.0],
  [0.0, 12.083045973594572, 12.083045973594572, 1.4142135623730951],
  [8.54400374531753,
   13.601470508735444,
   13.601470508735444,
   12.083045973594572]],
 'ivl': ['1', '0', '6', '9', '4', '8', '3', '2', '5', '7'],
 'leaves': [1, 0, 6, 9, 4, 8, 3, 2, 5, 7],
 'color_list': ['C1', 'C1', 'C1', 'C1', 'C1', 'C2', 'C2', 'C0', 'C0'],
 'leaves_color_list': ['C1',
  'C1',
  'C1',
  'C1',
  'C1',
  'C1',
  'C0',
  'C2',
  'C2',
  'C2']}
```




```
from sklearn.cluster import AgglomerativeClustering
```

```
clus = AgglomerativeClustering(n_clusters = 5)
```

```
c = clus.fit_predict(data2)
```

```
c
```

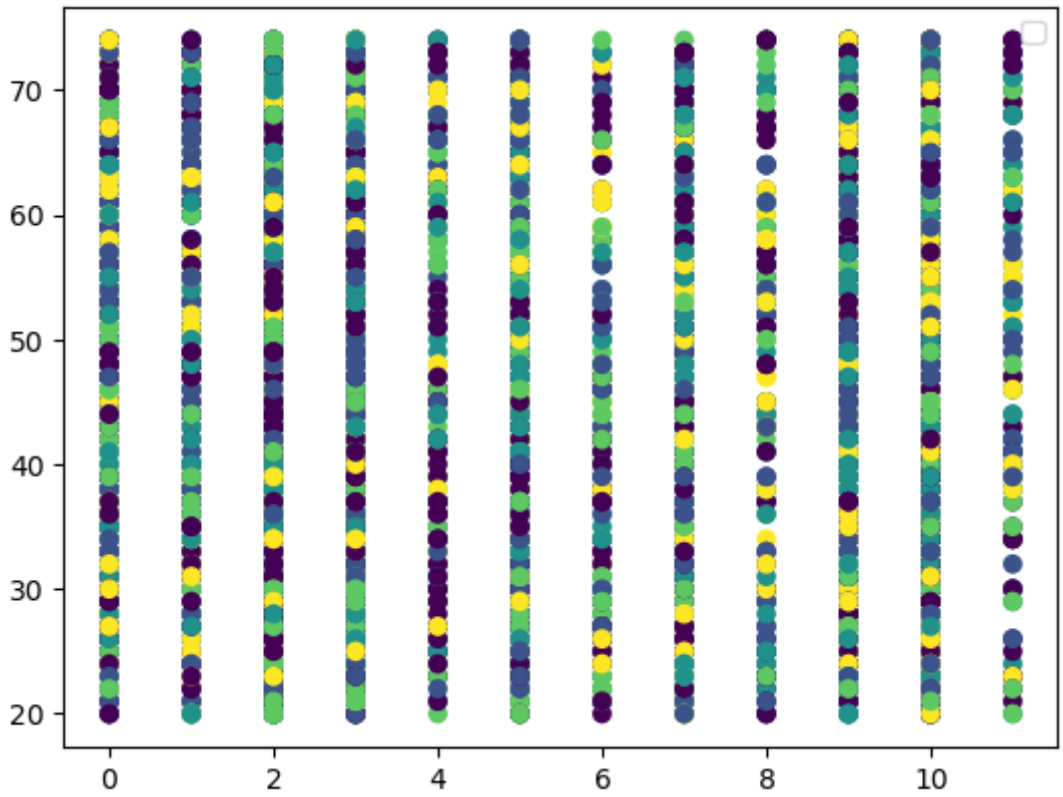
Out[68]:

```
array([2, 3, 0, ..., 0, 3, 3], dtype=int64)
```

In [70]:

```
plt.scatter(data2["Unit"],data2["growth_rate"], c = c)
plt.legend()
```

```
No artists with labels found to put in legend.  Note that artists whose la
bel start with an underscore are ignored when legend() is called with no a
rgument.
```

Out[70]:

```
<matplotlib.legend.Legend at 0x202428e87c0>
```